

Question Number	Answer	Additional Guidance	Mark
3(a)	There is no correlation between the percentage of visible gill remaining and the number of touches (1)	Candidates express this in several different ways Accept frequency (for number)	1

Question Number	Answer	Additional Guidance	Mark
3(b)	{bar / scatter} graph with linear scale (1) - complete axis labels (1) - all data plotted correctly (1)	MP1 Bar graph: y axis starts at zero, separate bars of same width Ignore lines in scatter graph Accept mean % gills visible and stimulus / (number of) touches 10 52 47 49 45 MP3 Do not accept if line extrapolated to origin	3

Question Number	Answer	Additional Guidance	Mark
3(c)(i)	correct completion of table (1)	0, 9, 0, 0, 9 Do not allow -9	1

Question Number	Answer	Additional Guidance	Mark
3(c)(ii)	- correct denominator used (1) - correct answer (1)	$120 / 5 \times 24 / 5(5^2-1)$ 0.1 Correct answer with no working gains two marks	2

Question Number	Answer	Additional Guidance	Mark
3(c)(iii)	- the calculated value of $r_s(0.100)$ is less than the critical value 1.000 (1) - therefore, accept the null hypothesis, so there is no correlation between the percentage response and the number of touches (1)	Allow ECF from 3cii if calculated value is more than stated critical value Accept critical value of 1 / 1.0 / 1.00 or marked in table at 5% significance	2

Question Number	Answer	Additional Guidance	Mark
3(d)	An answer that includes two of the following points: • use more than 5 touches (1) • change time intervals (between touches) (1) • use {more sea slugs / different ages / different species of sea slug} (1) • change a variable (related to sea water) (1)	Ignore increase sample size / repeat the experiment Accept temperature / pH / salinity / oxygen concentration / carbon dioxide concentration	2

Question Number	Answer	Additional Guidance	Mark
3(e)	An answer that includes one of the following points: • allows more oxygen uptake (1) • allows more feeding time (1) • (reduced response) conserves energy (1)		1

(Total for question 3 = 12 marks)

